

Francesco Romor

Curriculum Vitae

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Current position

November 2023– (Expected) October 2026 **Weierstrass Institute, Berlin, Germany**, *Postdoc*, Research group: Numerical Mathematics and Scientific Computing, Head: Prof. Volker John, Coworker: Dr. Alfonso Caiazzo.

Education

October 2019– October 2023 **International School for Advanced Studies (SISSA), Trieste, Italy**, *Ph.D. student*, Mathematical analysis, modelling and applications. Thesis title: **Nonlinear Parameter Space and Model Order Reductions enhanced by scientific machine learning**, Advisor: Prof. Gianluigi Rozza.
Defence: 29 September 2023

September 2017– October 2019 **University of Trieste, Trieste, Italy**, *Master's degree in mathematics*, 110/110 cum laude, Department of mathematics and geosciences, Thesis title: **Reduction in Parameter Space for Problems approximated by the Discontinuous-Galerkin Method in Computational Fluid Dynamics**. Advisor: Prof. Gianluigi Rozza

September 2014– September 2017 **University of Trieste, Trieste, Italy**, *Bachelor degree in mathematics*, 110/110 cum laude, Department of mathematics and geosciences, Thesis title: **"Geodesics on Lie Groups with invariant Metrics"**. Advisor: Prof. Giovanni Landi

Publications

Software Papers

2021 Francesco Romor, Marco Tezzele, and Gianluigi Rozza. **ATHENA: Advanced Techniques for High dimensional parameter spaces to Enhance Numerical Analysis**. *Software Impacts* 10 (2021): 100133.

Publications in Journals

- 2026 Francesco Romor, Felipe Galarce, Jan Brüning, Leonid Goubergrits, Alfonso Caiazzo. **Shape-Informed Graph Neural Networks and Data Assimilation: Application to Velocity and Pressure Reconstruction in Aortic Blood Flow**. *SIAM Journal on Imaging Sciences* 19 (2), 710-751
- 2026 Moaad Khamlich, Francesco Romor, and Gianluigi Rozza. **Efficient numerical strategies for entropy-regularized semi-discrete optimal transport**. *Computer Methods in Applied Mechanics and Engineering* 453, 118821
- 2025 Francesco Romor, Davide Torlo, and Gianluigi Rozza. **Friedrichs' systems discretized with the DGM: domain decomposable model order reduction and Graph Neural Networks approximating vanishing viscosity solutions**. *Journal of Computational Physics* (2025): 113915.
- 2025 Francesco Romor, Giovanni Stabile, and Gianluigi Rozza. **Explicable hyper-reduced order models on nonlinearly approximated solution manifolds of compressible and incompressible Navier-Stokes equations** *Journal of Computational Physics* 524 (2025): 113729.
- 2024 Guglielmo Padula, Francesco Romor, Giovanni Stabile, and Gianluigi Rozza. **Generative Models for the Deformation of Industrial Shapes with Linear Geometric Constraints: model order and parameter space reductions** *Computer Methods in Applied Mechanics and Engineering* 423 (2024): 116823.
- 2024 Francesco Romor, Marco Tezzele, Gianluigi Rozza. **A local approach to parameter space reduction for regression and classification tasks**. *Journal of Scientific Computing* 99, no. 3 (2024): 83.

- 2023 Francesco Romor, Giovanni Stabile, and Gianluigi Rozza. **Non-linear Manifold Reduced-Order Models with Convolutional Autoencoders and Reduced Over-Collocation Method**. Journal of Scientific Computing 94, 74 (2023). <https://doi.org/10.1007/s10915-023-02128-2>
- 2023 Francesco Romor, Marco Tezzele, Markus Mrosek, Carsten Othmer, and Gianluigi Rozza. **Multi-fidelity data fusion through parameter space reduction with applications to automotive engineering**. International Journal for Numerical Methods in Engineering, 124(23), 5293-5311. <https://doi.org/10.1002/nme.7349>
- 2022 Francesco Romor, Marco Tezzele, Andrea Lario, Gianluigi Rozza . **Kernel-based active subspaces with application to computational fluid dynamics parametric problems using the discontinuous Galerkin method**. International Journal for Numerical Methods in Engineering 123.23 (2022): 6000-6027.

Books

- 2022 Chapters 12 and 16: Gianluigi Rozza, Giovanni Stabile, and Francesco Ballarin, eds. **Advanced Reduced Order Methods and Applications in Computational Fluid Dynamics**. Society for Industrial and Applied Mathematics, 2022.

Preprints

- 2026 Sarah Katz, Francesco Romor, Jia-Jie Zhu, Alfonso Caiazzo **LDDMM stochastic interpolants: an application to domain uncertainty quantification in hemodynamics** (2026).
- 2025 Francesco Romor, Federico Pichi, Giovanni Stabile, Gianluigi Rozza, Christoph Schwab. **ROM for Viscous, Incompressible Flow in Polygons – exponential -width bounds and convergence rate** (2025).

Conference Proceedings

- 2021 Romor, Francesco, Marco Tezzele, and Gianluigi Rozza. **Multi-fidelity data fusion for the approximation of scalar functions with low intrinsic dimensionality through active subspaces**. In PAMM, vol. 20, no. S1, p. e202000349. Berlin: Wiley-VCH GmbH, 2021.

Contribution to national and international events

Visiting periods

- 2023 Visiting period at the Massachusetts Institute of Technology (MIT) for the MISTI MIT-Italy FVG Project, in collaboration with Prof. Youssef Marzouk, Boston, USA, 3-15 October 2023.

Speaker

- 2026 I will give an invited talk at the workshop **INdAM Workshop StructApp 2026 – Structure-preserving approximation of evolutive problems** 7-11 September 2026, Rome, Italy.
- 2026 I will give an invited talk at the workshop **BMS–BGSMath Junior Meeting 2026** 2-4 September 2026, Barcelona, Spain.
- 2026 I will give a contributed talk at the conference **WCCM-ECCOMAS 2026** 19-24 July 2026, in Munich, Germany.
- 2026 I will give a contributed talk at the conference **CMBE2026** 22-24 June 2026 in Kobe, Japan.
- 2026 I gave an invited talk at the conference **EMS TAG SciML 2026** at the University of Cologne, Germany, 10-12 March 2026.
- 2026 I gave a contributed talk at the conference **ML4FLUIDS2026** at CWI Amsterdam, Netherlands on the 4-6 March 2026.
- 2025 I gave a contributed talk at the conference **NOKO2025**, 17-18 October 2025, Berlin, Germany.
- 2025 I gave a contributed talk at the conference **COUPLED problems**, 25-28 May, 2025, Villasimius, Italy.
- 2025 I gave a contributed talk at **MESIGA Numerical Methods in Applied Mathematics**, 11-13 March, 2025, Potsdam, Germany.
- 2025 I gave a contributed talk at the **COLIBRI focus workshop on Computational Medicine**, 30-31 January, 2025, Graz, Austria.
- 2025 Invited speaker at the conference **DTE & AICOMAS 2025** with the talk on "Registration-based data assimilation from medical images", 17-21 February 2025, Paris, France.
- 2024 Invited speaker at the conference **ECCOMAS 2024 - "Efficient Numerical Resolution of Parametric Partial Differential Equations on Solution Manifolds Parametrized by Neural Networks"**, 3-7 June 2024, Lisbon, Portugal.

- 2023 Invited speaker at the conference ECCOMAS YIC 2023 - MS "Scientific Machine Learning techniques for complex engineering systems", 19-21 June 2023, Porto, Portugal
- 2023 Contributed talk at the conference SIAM CSE23, 27 February - 3 March 2023, Amsterdam, Netherlands
- 2022 Invited speaker at the conference GIMC SIMAI YOUNG, 29-30 September 2022, Pavia, Italy
- 2022 Contributed talk at Model Reduction and Surrogate Modeling (MORE), 19-23 September 2022, Berlin, Germany
- 2022 Contributed talk at the conference Young Mathematicians in Model Order Reduction - YMMOR, 18-22 July 2022, Munster, Germany

Workshops/Summer schools/Attended conferences

- 2024 I was poster presenter "Registration-based data assimilation of aortic blood flows" at the Leibniz MMS Days 2024, in Kaiserslautern, Germany.
- 2024 I attended the **workshop on Optimal transport** in Berlin, Germany, 11-15 March.
- 2023 I attended the "10th deal.II Users and Developers Workshop" at Leibniz University Hannover, Germany, September 11-15, 2023
- 2022 I was a monographic co-lecturer at the second edition of the Summer School on Reduced Order Methods in Computational Fluid Dynamics at SISSA
- 2021 I attended the International Conference in Coupled problems in Science and Engineering, 13-16 March 2021
- 2021 I attended the SIAM Conference on Computational Science and Engineering, Virtual Conference (originally scheduled in Fort Worth, Texas, U.S.) 1-5 March 2021.
- 2020 I was a poster presenter at the GAMM Juniors' Summer School on Applied Mathematics and Mechanics, July 2020
- 2019 I was a poster presenter at the Summer School on Reduced Order Methods in Computational Fluid Dynamics at SISSA.

Organization of minisymposia/seminars

- 2026 Minisymposium "Forward and inverse modelling in biomechanics" co-organizer with Felipe Galarce, Nicolás Barnafi, and Alfonso Caiazzo at the WCCM-ECCOMAS2026 conference.
- 2023 Seminar: "Efficiently solving parametric PDEs on solution manifolds parametrized by neural networks" at MIT, Boston, USA.
- 2021-2022 I organized with Sara Farinelli the weekly Analysis Junior Seminars at SISSA and the 2022 SISSA Women in mathematics colloquium.

Grants/Scholarships

- 2025-2027 Grant: **MATH+AA-Health-3: Randomization of Surrogate Models for the Quantification of Domain Uncertainty Propagation in Cardiovascular Models**. Project Heads: Alfonso Caiazzo, Ana Djurdjevac, Francesco Romor
- 2022 Grant: MISTI MIT-Italy FVG Project: multi-fidelity model order reduction for inverse problems at MIT and SISSA
- 2014-2017 Scholarship: I was given the scholarship offered by the Istituto Nazionale di Alta Matematica (INDAM) for the Bachelor degree in mathematics

Student supervision/Academic roles

- 2026 Master's thesis of Jannik Schlieffen in "**Nonparametric model order reduction for hemodynamics**", Master degree in Mathematics, Free University of Berlin
- 2022-2023 Master's thesis of Guglielmo Padula in "**Generative models for 3D object deformation with constraints applied to reduced order modelling**", Master's degree in Data Science, University of Trieste
- 2023 Master's thesis of Giorgio Abelli in "**Comparison of model order reduction methods for inverse problems enhanced by machine learning: FEM basis group equivariant layers on Graph Neural Networks**", Master degree in Data Science, University of Trieste
- 2021-2022 SISSA SIAM chapter president

Open-source software development

- SOT** I am the main developer of **SOT** a C++ library for entropy-regularized semi-discrete optimal transport and available on GitHub github.com/SemiDiscreteOT/SemiDiscreteOT
- ATHENA** I am the maintainer and main developer with Dr Marco Tezzele of **ATHENA** a python library for dimension reduction in parameter space and available on GitHub <https://github.com/mathLab/ATHENA>
- ITHACA-FV** I contributed to **ITHACA-FV** a C++ library for model reduction based on **OpenFOAM** and available on GitHub <https://github.com/mathLab/ITHACA-FV>

Relevant programming languages and software experience

Languages **C++, python**

Software **Numerical modelling of PDEs:** deal.II ([lifex-cfd](#), [exadg](#)), OpenFOAM, FEniCS

Software **Scientific machine learning:** PyTorch, JAX

Software **GPU computing:** Kokkos, cuda

Languages

Italian mother tongue

2014 First Certificate in English, Grade A, CEFR level C1

2025 German B2.1